CLOUD COMPUTING-II
UNIT-V
Cloud ML and AI

Introduction

Machine learning is programming computers to optimize a performance criterion using example data or Past experience. We have a model defined up to some parameters, and learning is the execution of a computer program to optimize the parameters of the model using the training data or experience. The model may be predictive to make predictions in the future, or descriptive to gain knowledge from data.

Machine learning is a subfield of artificial intelligence that involves the development of algorithms and statistical models that enable computers to improve their performance in tasks through experience. These algorithms and models are designed to learn from data and make predictions or decisions without explicit instructions. There are several types of machine learning, including supervised learning, unsupervised learning, and reinforcement learning.

Building ML and AI models on Cloud

Building Machine Learning (ML) and Artificial Intelligence (AI) models on the cloud involves several steps and practices. Here's a detailed guide on how to go about it:

1. Prepare your Data: Well-prepared data can accelerate model development. Ensure your data pipeline can process both batch and stream data to get consistent results from both types of data cloud.google.com.

2. Choose the Right Tools: Google Cloud provides several tools for building ML and AI models. One of the recommended tools is the Vertex AI Workbench notebooks for experimentation and development cloud.google.com.

3. Use Vertex AI SDK for Python: This Pythonic way allows you to use Vertex AI for your end-to-end model building workflows, which works seamlessly with popular ML frameworks including PyTorch, TensorFlow, XGBoost, and scikit-learn cloud.google.com.

4. Train a Model: Training a model within a notebook instance may be sufficient for small datasets or subsets of a larger dataset. For larger datasets or for distributed training, consider using the Vertex AI training service cloud.google.com.

5. Operationalize Job Execution with Training Pipelines: Create training pipelines to operationalize training job execution on Vertex AI. A training pipeline encapsulates training jobs cloud.google.com.

6. Orchestrate the ML Workflow: Use Vertex AI Pipelines to orchestrate the flow. This includes data process, training, evaluation, test, and deployment cloud.google.com.

7. Monitor Prediction Performance: Monitoring tools should be used to track model performance and respond to changes. Models that provided excellent performance at first can degrade in performance over time due to changes to data inputs run.ai.

8. Deploy the Model: Choose hardware that is appropriate for your model, like different central processing unit (CPU) virtual machine (VM) types or graphics processing unit (GPU) types cloud.google.com.

9. Use a Source Control Repository: Use source control to version control your ML pipelines and the custom components you build for those pipelines. Use Artifact Registry to store, manage, and secure your Docker container images without making them publicly visible cloud.google.com.

10. Fine Tune Alert Thresholds: Tune the thresholds used for alerting so you know when skew or drift occurs in your data. Alert thresholds are determined by the use case, the user's domain expertise, and by initial model monitoring metrics cloud.google.com.

11. Regularly Compute New Feature Values: Often, a model will use a subset of features sourced from Vertex AI Feature Store. Regularly compute the new feature values at the required cadence, depending upon feature freshness needs, and ingest them into Vertex AI Feature Store for online or batch serving cloud.google.com.

12. Use Pre-Tuned AI Services: Cloud machine learning platforms provide optimized AI services for use cases like computer vision, natural language processing, speech synthesis, and predictive analytics. These services are typically trained and tested using more data than is available to most businesses run.ai.

ML and GCP

The process of implementing Machine Learning (ML) and deploying it on Google Cloud Platform (GCP) involves several steps. Here's a step-by-step guide on how to do this:

13. Prepare Your Data:

Before you can start implementing machine learning, you need to have a large set of training data that includes the attribute you want to predict based on other features cloud.google.com. This data should be stored in a Google Cloud Storage bucket cloud.google.com.

2. Use Vertex AI Workbench for Experimentation and Development:

Vertex AI Workbench notebooks are recommended for experimentation and development, including writing code, starting jobs, running queries, and checking status cloud.google.com.

3. Train Your Model:

After preparing your data, you can start training your model. You can use Google AI Platform for this which is a one-stop solution for machine learning developers and data scientists to take their ML projects from experiment to production and deployment medium.com.

Cloud ML and AI

4. Evaluate Model Accuracy and Tune Hyperparameters:

After training your model, evaluate its accuracy. If necessary, tune your model's hyperparameters to improve its performance cloud.google.com.

5. Deploy Your Model:

Once your model is trained and evaluated, you can deploy it. There are several ways to do this on GCP, including Google App Engine and Google Cloud Functions developer.nvidia.com. For example, to deploy your model on Google App Engine, you would need to create a web service using Flask, upload your trained model to Google Cloud Storage, and then deploy your application using the gcloud app deploy command developer.nvidia.com.

6. Monitor Your Model:

After deploying your model, you need to monitor its performance to ensure that it's performing as expected. Vertex AI provides tools for skew detection and drift detection cloud.google.com.

7. Regularly Compute New Feature Values:

If your model uses a subset of features from Vertex AI Feature Store, you should regularly compute the new feature values and ingest them into Vertex AI Feature Store for online or batch serving cloud.google.com.This is a high-level overview of the process. Each of these steps involves more detailed work, and you may need to iterate and refine your model as you progress through the process.

Cloud AutoML

Cloud AutoML, or Automated Machine Learning, is a suite of machine learning products that enables developers with limited machine learning expertise to train high- quality models specific to their business needs. It leverages state-of-the-art transfer learning and Neural Architecture Search technology to make this process
easier pypi.org.

AutoML can be used across various data types and model objectives. Google Cloud's AutoML solutions include AutoML Vision, which is used for image recognition, AutoML Natural Language, for text analysis, and AutoML Video, for video data analysis. Each of these solutions helps in creating custom machine learning models for specific use cases like object detection, classification, sentiment analysis, action recognition, and more cloud.google.com.

To use AutoML, you need to provide the training data. For image data, you can use AutoML to train an ML model to classify image data or find objects in image data. For text data, AutoML uses machine learning to analyze the structure and meaning of text data. You can use AutoML to train an ML model to classify text data, extract information, or understand the sentiment of the authors. For video data, AutoML uses machine learning to analyze video data to classify shots and segments, or to detect and track multiple objects in your video data cloud.google.com.

AutoML also supports ensemble models, which are enabled by default. Ensemble learning improves machine learning results and predictive performance by combining multiple models as opposed to using single models. The ensemble iterations appear as the final iterations of your job learn.microsoft.com.

To install the AutoML API Python Client, you can follow the instructions provided in pypi.org. Once installed, you can use the API to interact with the AutoML service. For example, you can use AutoMLCreateDatasetOperator to create a dataset, AutoMLImportDataOperator to import data into the dataset, and AutoMLListDatasetOperator to get a list of datasets airflow.apache.org.

In conclusion, AutoML offers a powerful and accessible way to create custom machine learning models for various data types and use cases, without requiring extensive machine learning expertise. It simplifies the process of training and deploying models, making it a valuable tool for both developers and businesses.

Google’s pre-trained ML APIs

Google offers a variety of pre-trained Machine Learning (ML) APIs that can be used for a wide range of applications. These APIs can be used for tasks such as image recognition, natural language processing, speech recognition, video analysis, and more.

One of the key resources provided by Google for ML is the Model Garden on Vertex AI. This platform provides a curated collection of over 100 models, including enterprise-ready foundation model APIs, open-source models, and task-specific models from Google and third-party providers cloud.google.com.

Some of the APIs available in Google's ecosystem include:

Cloud Video Intelligence:

This API allows users to understand their videos at the shot, frame, or video level. It can detect labels, provide shot change detection, and specify the region where video API requests should be executed ryanwingate.com.

Cloud Speech:

This API enables speech to text transcription in over 100 languages. It takes an audio file as input and returns a text transcript. It also supports speech timestamps that provide start and end times for every word in the audio transcription ryanwingate.com.

Cloud Natural Language:

This API enables developers to understand text with a single API request.

It performs several distinct functions such as detecting sentiment, analyzing syntax, and classifying content ryanwingate.com.

Vision API:

This API, part of the ML Kit, allows developers to detect and extract information about entities in an image across a broad group of categories. The default model provided with ML Kit recognizes more than 400 different entities developers.google.com.

Here is an example of how to use the Vision API:

```
from google.cloud import vision

def detect_text(path):
    """Detects text in the file."""
    client =
    vision.ImageAnnotatorClient
    ()

    with io.open(path, 'rb') as image_file:
        content = image_file.read()

    image =
    vision.Image(content=content)

    response = client.text_detection(image=image)
    texts = response.text_annotations
    print('Texts:')

    for text in texts: print('\
        n"{}"'.format(text.description))

        vertices = (['({},{})'.format(vertex.x,
        vertex.y)
                    for vertex in
                    text.bounding_poly.vertices])

        print('bounds:
        {}'.format(','.join(vertices)))
```

This code will detect and print the text in an image file ryanwingate.com.

Google also provides tools for creating custom models using TensorFlow, an open-source library provided by the Google Brain team. These models can be trained using your own data and can be deployed on managed Google infrastructure using the Machine Learning Engine ryanwingate.com.

Cloud ML and AI

To implement a model, it's recommended to start simple and gradually iterate on creating a better model. Simple models provide a good baseline, even if you don't end up launching them. In fact, using a simple model is probably better than you think. Starting simple helps you determine whether or not a complex model is even justified developers.google.com.

Remember to monitor your model and set up alerts for training-serving skew, which occurs when the incoming features used for inference have values that fall outside the distribution range of the data used in training developers.google.com.

------------ ALL THE BEST STUDENTS---------